Measuring Consistency in LLM-Based Resume-To-Job Matching

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Abstract

Employing artificial intelligence models in the staffing process is complicated, especially when committed to ethical practices, such as transparency and impartiality. To evaluate this premise, the consistency of general models will be followed throughout the resume-to-job matching process. The dataset consists of synthetic job postings and resumes from Hugging Face. Original resumes were compared against a slightly altered version, missing the first sentence of the summary. Three lightweight, open-source models were selected from the Hugging Face Sentence Transformer module for this test. These models were used to embed the job posting, original resume, and modified resume. The resumes were then scored based on their similarity to the job posting; the difference between these two scores was used to evaluate the models. Although all models had high variance, `all-MiniLM-L6-v2` performed the best, with a substantially lower flag rate. In resume-to-job scoring, all models had a potential for high variability and often disagreed within the same matching task. Blindly applying AI to a field that is inherently people-oriented, such as staffing, can be detrimental. Thoroughly vetted commercial or custom models would be better suited to resume-to-job matching.

1 Introduction & Motivation

With the rise of artificial intelligence, there has been a rush to adopt the new technology to streamline human-intensive tasks, such as staffing. However, AI adoption is not as straightforward as selecting a model and handing it resumes and a job posting. To test this theory, the consistency and sensitivity of large language models will be evaluated through the resume-to-job matching process. This will be accomplished by comparing the original resume scoring to the scoring of a slightly altered version of the resume. The evaluation will attempt to pinpoint the potential pitfalls of blindly applying general LLMs to the complicated task of staffing. Additionally, the assessment emphasizes the importance of ethical AI use in staffing, highlighting the need for fairness and explainability.

2 Dataset

For the experiment, the dataset consists of synthetic job postings and resumes obtained through Hugging Face. The job postings are in CSV format. Information, like company name, position title, and job description, for each job posting is contained in a spreadsheet. The resumes are in JSONL format, where each resume is a JSON object separated by a newline. Additionally, each resume was subtly altered for comparison by removing the first sentence of the summary. The resume alterations were chosen for simplicity since it was a straightforward operation considering the original format. In addition, the slight change should result in fairly consistent model results regarding resume-to-job scoring; however, this was not always the case.

3 Methodology

3.1 Model Selection

Three open-source large language models were selected for initial testing to simplify the experiment and to avoid proprietary bias. The models are from Hugging Face's Sentence Transformers module, and they were selected since they are popular and lightweight. The models used in the tests were `all-MiniLM-L6-v2`, `all-MiniLM-L12-v2`, and `paraphrase-MiniLM-L6-v2`. `all-MiniLM-L6-v2` features 22.7 million parameters, while `all-MiniLM-L12-v2` features 33.7 million parameters; although `paraphrase-MiniLM-L6-v2` has the same number of parameters as `all-MiniLM-L6-v2`, it is specialized in paraphrase mining and semantic search. Each model was prompted identically, with the resume and job posting description embedded by the model and then compared.

3.2 Scoring Procedure

Resume and job matching were conducted through embedding and scoring the embeddings based on their similarity; this produced a score and rank for each resume given a job posting. Additionally, the altered resume was scored and ranked during this process, allowing the consistency of the model to be examined. The consistency was quantified by the delta, which is the difference between the original resume score and the altered resume score. Although this is used as an in-model comparison, it is also used for cross-model comparison, allowing the overall consistency of the model to be assessed.

3.3 Evaluation Metrics

There are two categories of evaluation: resume-to-job scoring and model statistics centered on the delta of the original resume and the modified resume. Cosine similarity and ranking are used to evaluate resume-to-job scoring; this is used to compare initial versus altered resumes, as well as to visualize the top candidate shift for a job posting, among others. Variance is used to compare model consistency and sensitivity to modifying resumes. In addition to variance, standard deviation, mean, minimum value, and maximum value were considered when assessing the delta scores.

3.4 Pipeline Architecture

The experiments were conducted using the notebook `notebooks/model_analysis.ipynb`. Each resume and altered resume were scored for each job posting using the cosine similarity of their embeddings; this process was replicated for each model. After completion, the results were exported to `data/results.csv`. The results feature columns identifying the job ID, resume ID, baseline score, variant score, delta, flag, and model name; the flag field was used to quantize the delta values.

3.5 Output & Visualization

The visualization of results is performed primarily through the associated application of the project. The app allows for several plots and tables depending on the selection of job posting, resume, and model; more information about app functionality can be found in `docs/App_Documentation.md`. Resume-to-job scoring is displayed using bar charts and similar tables; this depicts score variance between similar resumes as well as cross-model consistency. Model statistics are displayed in pie charts and tables; the charts use the flag field to narrow the delta scores into three categories, while a table is used to show additional metrics.

4 Results

4.1 Model Statistics

The difference between original resume scores and altered resume scores was used as the primary performance indicator for the three models. All models had a high variance. However, `all-MiniLM-L6-v2` was noticeably more stable than the other models. The other two models had a significantly higher flag rate, with many altered resume scores deviating more than 5% from the original resume score. Additionally, the minimum and maximum delta scores make it evident that `all-MiniLM-L12-v2` and `paraphrase-MiniLM-L6-v2` are more volatile than `all-MiniLM-L6-v2`; this is seen in both directions depending on the resume. Despite this observation in the other two models, `all-MiniLM-L6-v2` also has the tendency for the altered resume score to diverge greatly from the initial score.

4.2 Resume-To-Job Scoring

The similarity score between resumes and job posting was used as the performance measure for resume-to-job scoring; raw resume ranking was also used for evaluation. Once again, there was high variability in the original score versus the altered score. After removing the first sentence of the resume's summary, the scores were observed to increase, decrease, or stay the same. Even when examining the same resume and job posting, each model can respond differently to the changes in the resume. This behavior suggests that general models are not well-suited to staffing tasks, like matching many candidate resumes to a company's job posting.

Figure 1: Resume-to-job scoring of `resume_0` and `job_0`

Table 1: Model Statistics

Model	var	std	mean	min	max
all-MiniLM-L6-v2	0.001074	0.032768	0.013046	-0.144222	0.239736
all-MiniLM-L12-v2	0.001999	0.044714	0.042131	-0.125257	0.301305
paraphrase-MiniLM-L6-v2	0.003635	0.060289	0.038434	-0.190418	0.495978

Table 2: Model Flag Rate

Model	No Flag	Soft Flag	Hard Flag
all-MiniLM-L6-v2	85.30%	14.47%	0.23%
all-MiniLM-L12-v2	61.05%	36.30%	2.65%
paraphrase-MiniLM-L6-v2	58.71%	37.87%	3.43%

[1] $No\ Flag\ = \Delta < 5\%$
[2] $Soft\ Flag\ = 5\% \leq \Delta < 15\%$
[3] $Hard\ Flag\ = \Delta \geq 15\%$

4.3 Top Candidate Shift

Much like the resume-to-job scoring, the top candidate shift utilizes a similarity score and overall ranking; in this case, the top ten candidates were considered. The top candidate score typically decreases or stays the same; in rare cases, the score can increase. The common score shift suggests that the models are over-reliant on the summary, despite being a small part of the resume. It is possible that the entire resume cannot fit into the model's context window, which would emphasize the contents of the summary. In addition, the models could be less sensitive to itemized lists later in the resume as compared to the sentence structure of the summary. Again, specialized models may be required to create a reliable staffing solution.

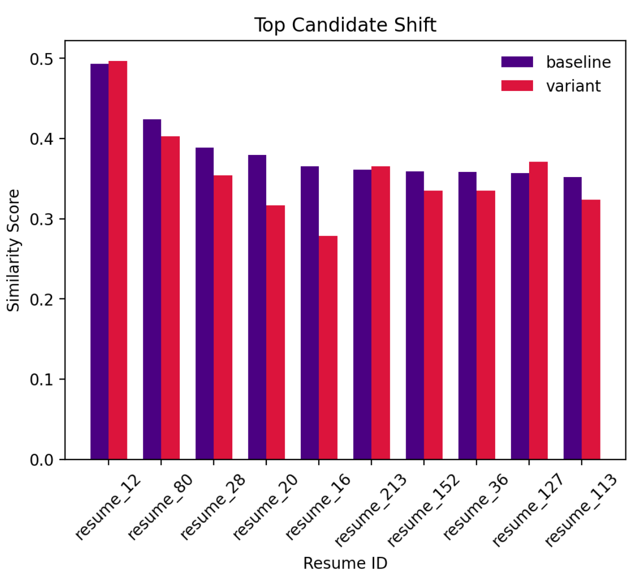

Figure 2: Top candidate shift of `job_0` with `all-MiniLM-L6-v2`

5 Conclusion & Future Work

Despite the surge in AI adoption to accelerate tasks, like staffing, employing new technologies requires careful consideration to retain integrity and clarity. It is evident that lightweight general-purpose models struggle with resume-to-job matching, with all models displaying high variance. Additionally, the models responded differently to the same task; this indicates the need for specialized models tailored to specific staffing departments. The models tested in this experiment are inherently limited due to their lightweightness, which can be seen in their constrained context windows. The limitations encountered suggest that stronger, specialized models are more appropriate when matching resumes to job postings.

Since small, generalized models are not sufficient for resume-to-job matching, other avenues should be explored. Commercial models from OpenAI, Google, or Anthropic could be easily incorporated into the existing resume-to-job matching process to test the utility of stronger models. Alternatively, a specialized model could be built using a pretrained model as a foundation. A model dedicated to matching resumes to job postings could be created through fine-tuning or retrieval augmented generation; both approaches would be viable with different advantages. Utilizing either a commercial model or creating a specialized model for resume-to-job matching would improve upon the lightweight models, addressing many of their shortcomings.